

EXPERIMENT 16

Determination and plotting of Radiation Pattern for a uniform linear array (ULA).

Aim:

To determine and plot the Radiation pattern by Simulating the array factor for a uniform linear array (ULA) using the given parameters and visualize the normalized array factor in polar form.

Software required:

- SCILAB version 6.0.1

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1 clear;
2 clear;
3 clf();
4
5 // =====
6 // PARAMETERS
7 // =====
8
9 frequency = 2.4e9; // Frequency = 2.4 GHz
10 c = 3e8; // Speed of light = 3 x 10^8 m/s
11
12 // Wavelength
13 lambda = c / frequency;
14
15 // Element spacing
16 d = lambda / 2;
17
18 // Number of antenna elements
19 N = 8;
20
21 // Phase difference
22 beta = 0;
23
24 // Angular range: 0 to 2*pi
25 theta = linspace(0, 2 * %pi, 720);
26
27 // =====
28 // ARRAY FACTOR
29 // =====
30
31 // Wave number
32 k = 2 * %pi / lambda;
33
34 // Phase term
35 psi = k * d * sin(theta) + beta;
36
37 // Initialise Array Factor
38 AF = zeros(1, length(theta));
39
40 // Calculate Array Factor
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?



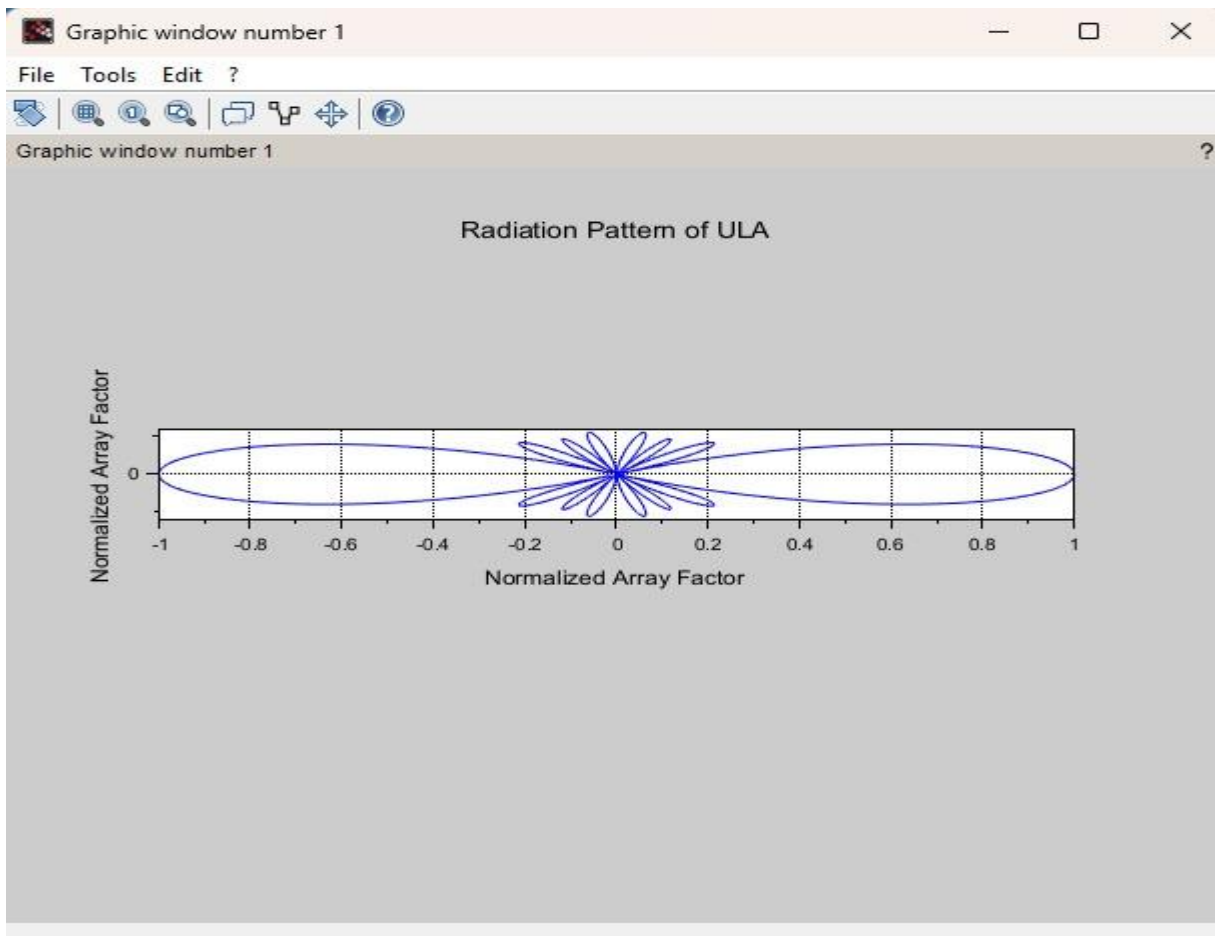
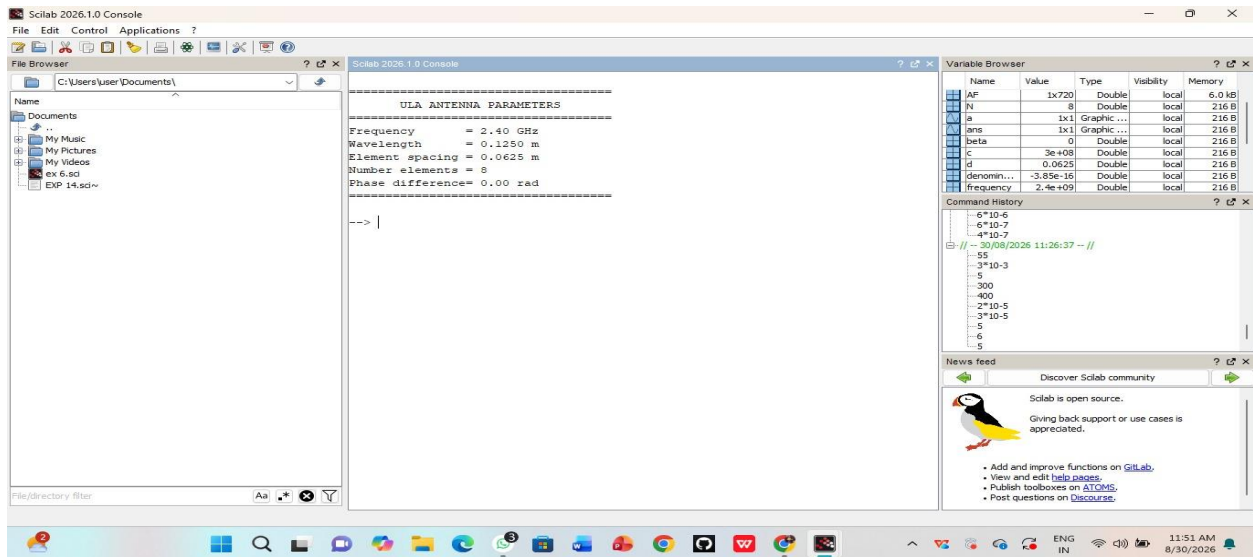
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci X

```
40
41 //Calculate Array Factor
42 for i = 1:length(theta)
43
44     denominator = sin(psi(i)/2);
45
46     //Avoid division by zero
47     if abs(denominator) < 1e-10
48         AF(i) = 1;
49     else
50         AF(i) = abs(sin(N * psi(i)/2) / (N * denominator));
51     end
52 end
53
54 end
55
56
57 //=====
58 //NORMALIZATION
59 //=====
60
61 AF = AF / max(AF);
62
63
64 //=====
65 //CONVERT POLAR TO CARTESIAN
66 //=====
67
68 x = AF .* cos(theta);
69 y = AF .* sin(theta);
70
71
72 //=====
73 //RADIATION PATTERN
74 //=====
75
76 figure(1);
77
78 plot(x, y);
79
80 a = gca();
81 a.isoview = "on";
```

```
79
80 a = gca();
81 a.isoview = "on";
82
83 xlabel("Normalised Array Factor");
84 ylabel("Normalised Array Factor");
85
86 title("Radiation Pattern of ULA");
87
88 xgrid();
89
90
91 //=====
92 //DISPLAY PARAMETERS
93 //=====
94
95 mprintf("\n");
96 mprintf("===== \n");
97 mprintf("----- ULA ANTENNA PARAMETERS ----- \n");
98 mprintf("===== \n");
99 mprintf("Frequency ----- = %.2f GHz \n", frequency / 1e9);
100 mprintf("Wavelength ----- = %.4f m \n", lambda);
101 mprintf("Element spacing = %.4f m \n", d);
102 mprintf("Number elements = %d \n", N);
103 mprintf("Phase difference = %.2f rad \n", beta);
104 mprintf("===== \n");
105
```

Output:



Case Study 1: Optimization of Beamwidth for High-Resolution RADAR Systems

Program:

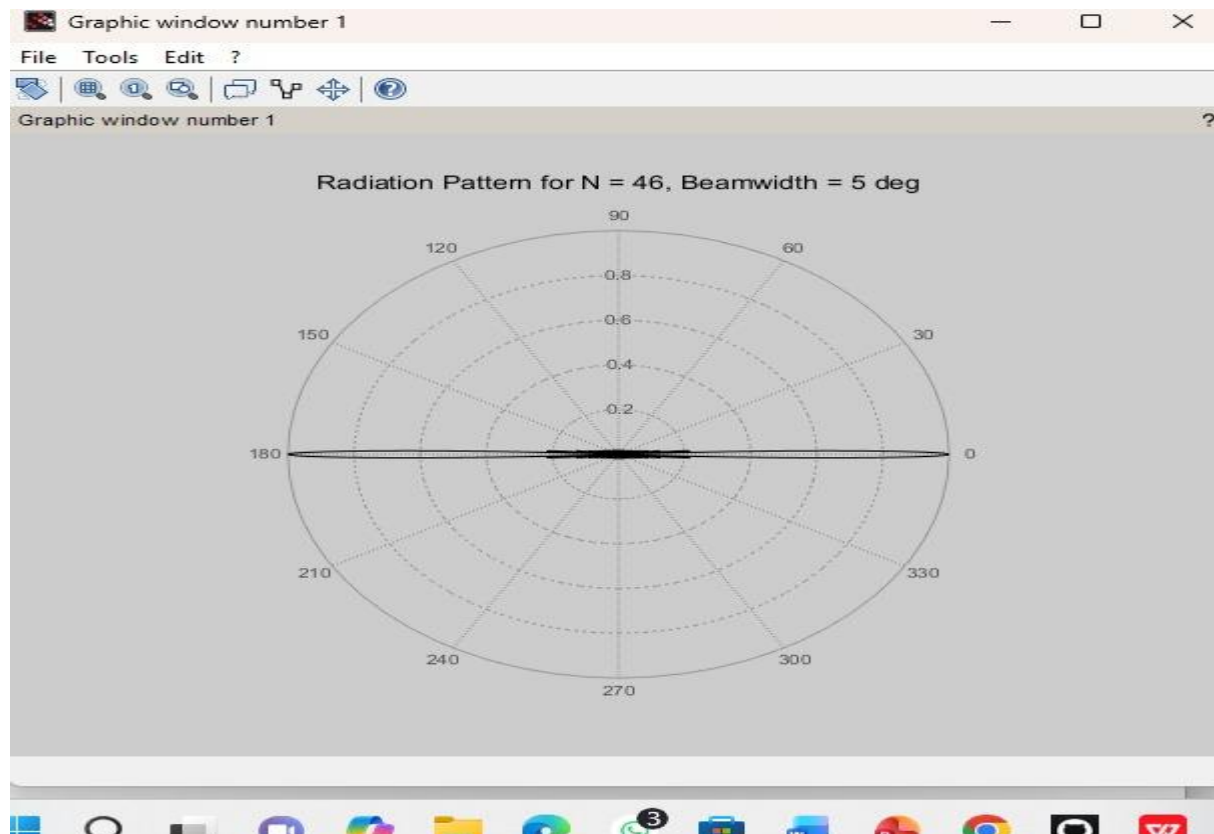
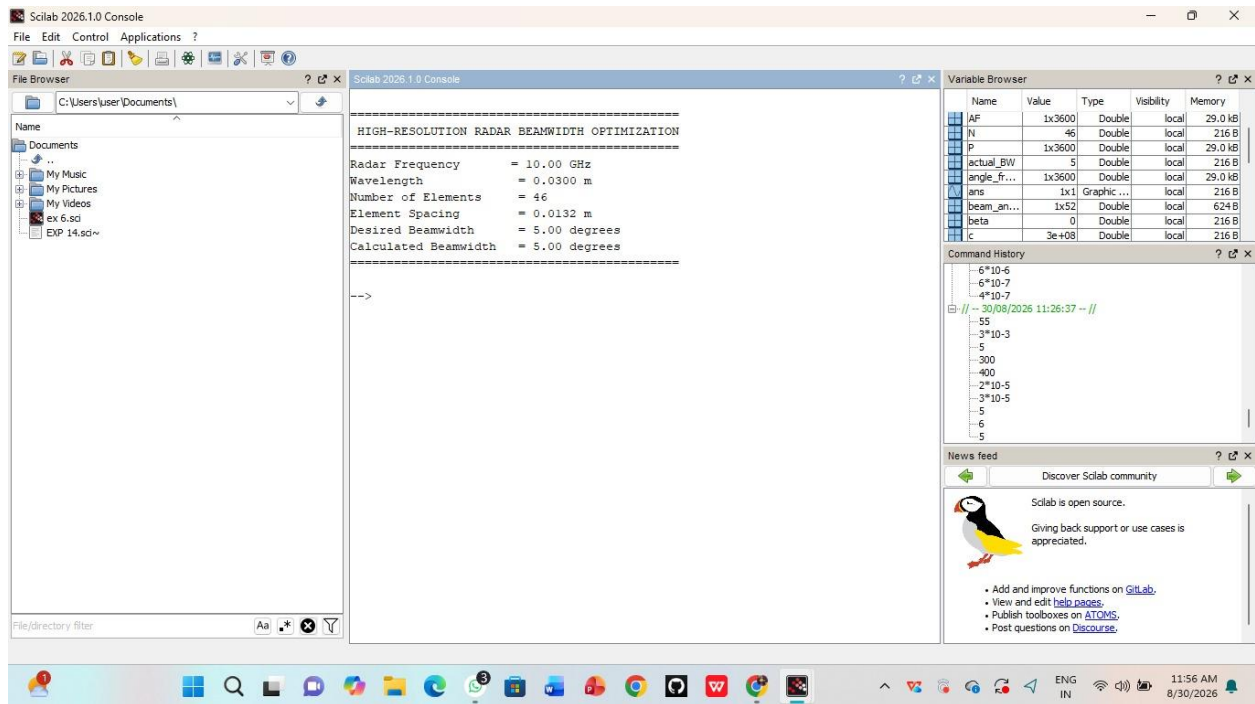
```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1 clc;
2 clear;
3 clf();
4
5 //=====
6 // OPTIMIZATION OF BEAMWIDTH FOR HIGH-RESOLUTION RADAR
7 //=====
8
9 // Radar parameters
10 frequency = 10e9; // Frequency = 10 GHz
11 c = 3e8; // Speed of light (m/s)
12 lambda = c / frequency; // Wavelength (m)
13
14 // Antenna array parameters
15 N = 46; // Number of elements
16 desired_BW = 5; // Desired beamwidth in degrees
17 beta = 0; // Phase difference
18
19 // Element spacing selected for approximately 5-degree HPBW
20 d = 0.44 * lambda;
21
22 // Angular range
23 theta = linspace(0, 2 * pi, 3600);
24
25 // Wave number
26 k = 2 * pi / lambda;
27
28 //=====
29 // ARRAY FACTOR CALCULATION
30 //=====
31
32 psi = k * d * sin(theta) + beta;
33
34 AF = zeros(1, length(theta));
35
36 for i = 1:length(theta)
37     denominator = sin(psi(i) / 2);
38     if abs(denominator) < 1e-10 then
39         AF(i) = 1;
40     end
41 end
```

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

82 actual_BW = desired_BW;
83 end
84
85 //=====
86 // CONVERT POLAR COORDINATES
87 //=====
88
89 x = AF .* cos(theta);
90 y = AF .* sin(theta);
91
92 //=====
93 // POLAR RADIATION PATTERN
94 //=====
95
96 figure(1);
97
98 subplot(theta, AF);
99
100 title("Radiation Pattern for N=46, Beamwidth=5-deg");
101
102 //=====
103 // DISPLAY RESULTS
104 //=====
105
106 fprintf("\n");
107 fprintf("=====");
108 fprintf("HIGH-RESOLUTION RADAR BEAMWIDTH OPTIMIZATION\n");
109 fprintf("=====");
110
111 fprintf("Radar Frequency = %.2f GHz\n", frequency / 1e9);
112 fprintf("Wavelength = %.4f m\n", lambda);
113 fprintf("Number of Elements = %d\n", N);
114 fprintf("Element Spacing = %.4f m\n", d);
115 fprintf("Desired Beamwidth = %.2f degrees\n", desired_BW);
116 fprintf("Calculated Beamwidth = %.2f degrees\n", actual_BW);
117
118 fprintf("=====");
119
```

Output:



Case Study 2: End-Fire Radiation for Wireless Communications

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1  clc;
2  clear;
3  clf();
4
5  // =====
6  // END-FIRE AND BROADSIDE RADIATION PATTERN
7  // FOR WIRELESS COMMUNICATION
8  // =====
9
10 // Basic parameters
11 frequency = 2.4e9; // Frequency = 2.4 GHz
12 c = 3e8; // Speed of light (m/s)
13 lambda = c / frequency; // Wavelength (m)
14
15 // Antenna array parameters
16 N = 8; // Number of elements
17 d = lambda / 2; // Element spacing = lambda/2
18
19 // Angular range
20 theta = linspace(0, 2 * %pi, 3600);
21
22 // Wave number
23 k = 2 * %pi / lambda;
24
25 // =====
26 // FUNCTION TO CALCULATE ARRAY FACTOR
27 // =====
28
29 function AF = array_factor(N, k, d, theta, beta)
30
31     psi = k * d * cos(theta) + beta;
32
33     AF = zeros(1, length(theta));
34
35     for i = 1:length(theta)
36
37         denominator = sin(psi(i) / 2);
38
39         if abs(denominator) < 1e-10 then
40             AF(i) = N;
41         else
42             AF(i) = sin(N * denominator) / denominator;
43         end
44     end
45
46 // =====
47 // DISPLAY PARAMETERS
48 // =====
49
50 fprintf("\n");
51 fprintf("===== \n");
52 fprintf("--- ANTENNA ARRAY RADIATION PATTERN \n");
53 fprintf("Frequency ----- %0.2f GHz \n", frequency / 1e9);
54 fprintf("Wavelength ----- %0.2f m \n", lambda);
55 fprintf("Number of Elements = %d \n", N);
56 fprintf("Element Spacing = %0.4f m \n", d);
57 fprintf("End-Fire Beta = %0.2f rad \n", beta);
58 fprintf("----- \n");
59
60 subplot(1, 2, 1);
61 plot(x_broadside, y_broadside);
62 a = gca();
63 a.isoview = "on";
64 axisid();
65
66 xlabel("Normalised Pattern");
67 ylabel("Normalised Pattern");
68
69 title("Broadside Configuration (beta = 0)");
70
71 // Draw circular reference rings
72 for r = [0.2 0.4 0.6 0.8 1.0]
73
74     xr = r * cos(theta);
75     yr = r * sin(theta);
76
77     plot(xr, yr);
78 end
79
80 // =====
81 // DISPLAY PARAMETERS
82 // =====
83
84 fprintf("\n");
85 fprintf("===== \n");
86 fprintf("--- ANTENNA ARRAY RADIATION PATTERN \n");
87 fprintf("Frequency ----- %0.2f GHz \n", frequency / 1e9);
88 fprintf("Wavelength ----- %0.2f m \n", lambda);
89 fprintf("Number of Elements = %d \n", N);
90 fprintf("Element Spacing = %0.4f m \n", d);
91 fprintf("End-Fire Beta = %0.2f rad \n", beta);
92 fprintf("----- \n");
```


EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?



EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci

```
84
85
86 //=====
87 // PLOT BOTH PATTERNS
88 //=====
89
90 figure(1);
91
92 //----- END-FIRE -----
93
94 subplot(1,2,1);
95
96 plot(x_endfire, y_endfire);
97 a = gca();
98 a.isoview = "on";
99
100 xgrid();
101
102 xlabel("Normalised Pattern");
103 ylabel("Normalised Pattern");
104
105 title("End-Fire Configuration (beta = pi/2)");
106
107
108 // Draw circular reference rings
109 for r = [0.2 0.4 0.6 0.8 1.0]
110
111     xr = r * cos(theta);
112     yr = r * sin(theta);
113
114     plot(xr, yr);
115
116 end
117
118
119 //----- BROADSIDE -----
120
121 subplot(1,2,2);
122
123 plot(x_broadside, y_broadside);
124 a = gca();
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

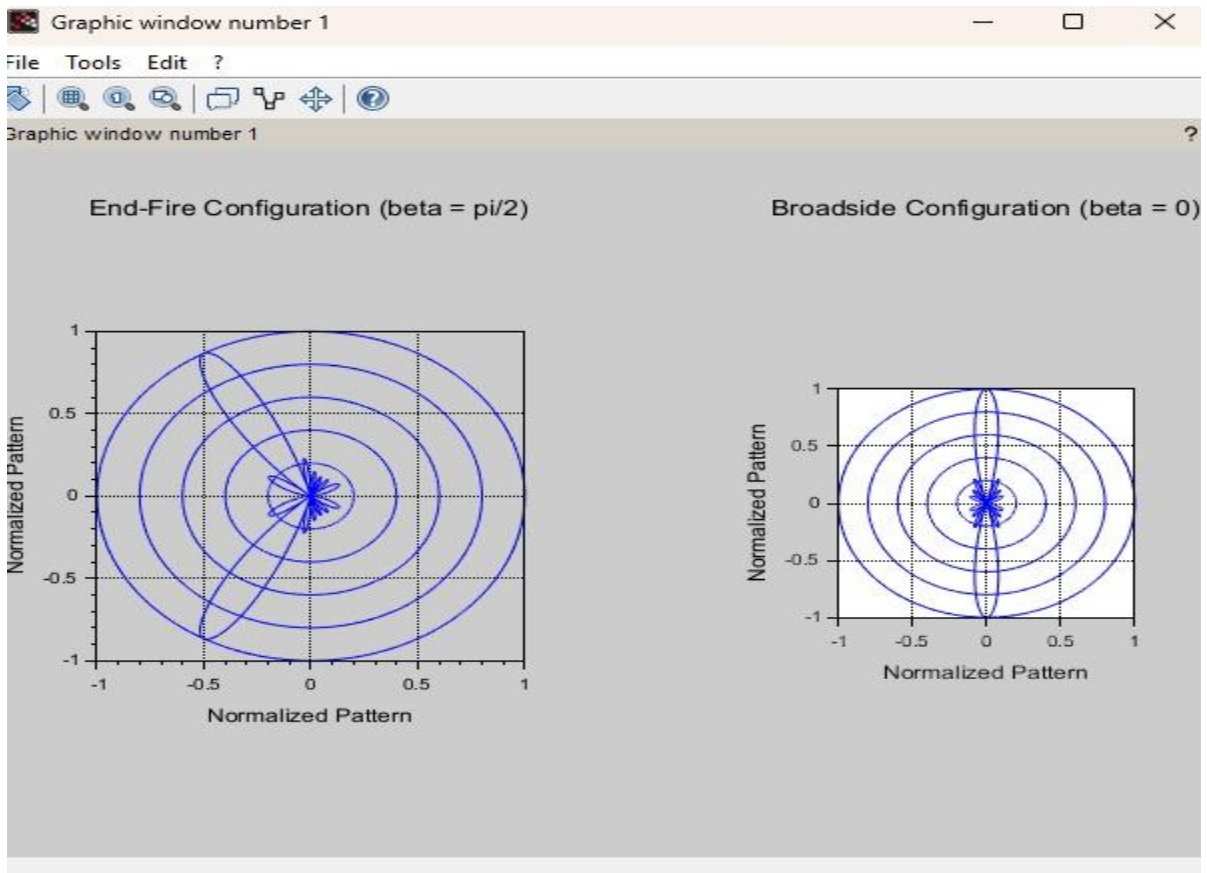
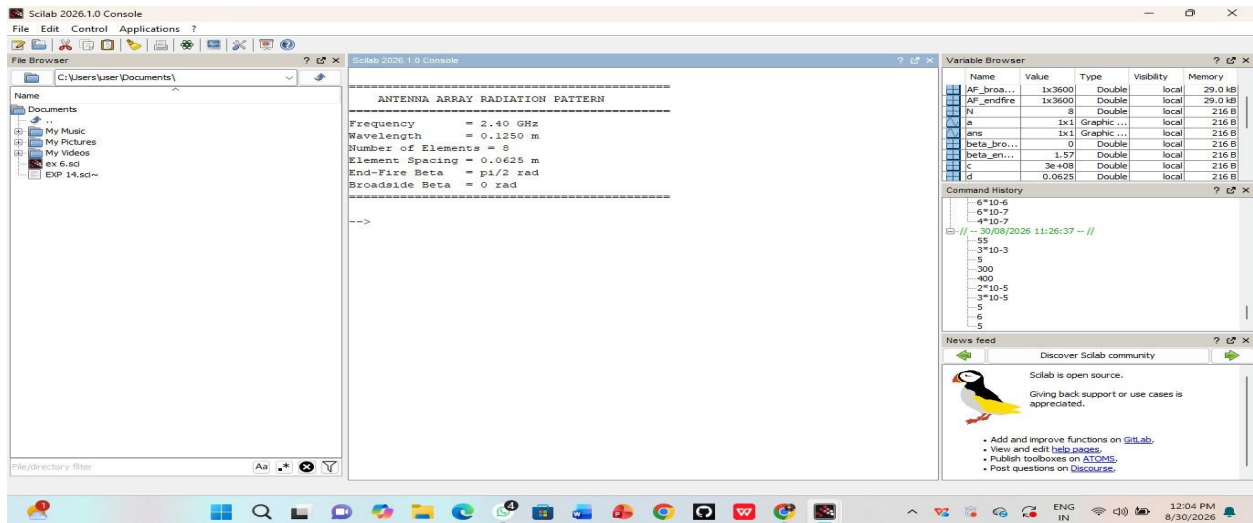


EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci

```
20
21 AF = AF / max(AF);
22
23 endfunction
24
25
26 //=====
27 // 1. END-FIRE CONFIGURATION
28 // beta = pi/2
29 //=====
30
31 beta_endfire = %pi / 2;
32
33 AF_endfire = array_factor(N, k, d, theta, beta_endfire);
34
35
36 //=====
37 // 2. BROADSIDE CONFIGURATION
38 // beta = 0
39 //=====
40
41 beta_broadside = 0;
42
43 AF_broadside = array_factor(N, k, d, theta, beta_broadside);
44
45
46 //=====
47 // POLAR TO CARTESIAN CONVERSION
48 //=====
49
50 x_endfire = AF_endfire .* cos(theta);
51 y_endfire = AF_endfire .* sin(theta);
52
53 x_broadside = AF_broadside .* cos(theta);
54 y_broadside = AF_broadside .* sin(theta);
55
56
57 //=====
58 // PLOT BOTH PATTERNS
59 //=====
```

Output:



RESULT: The Experiment was done and the radiation characteristics of the ULA is plotted. The different case studies were studied and analysed.